The weird and wonderful world of CJK fonts

A borderline non-programming talk



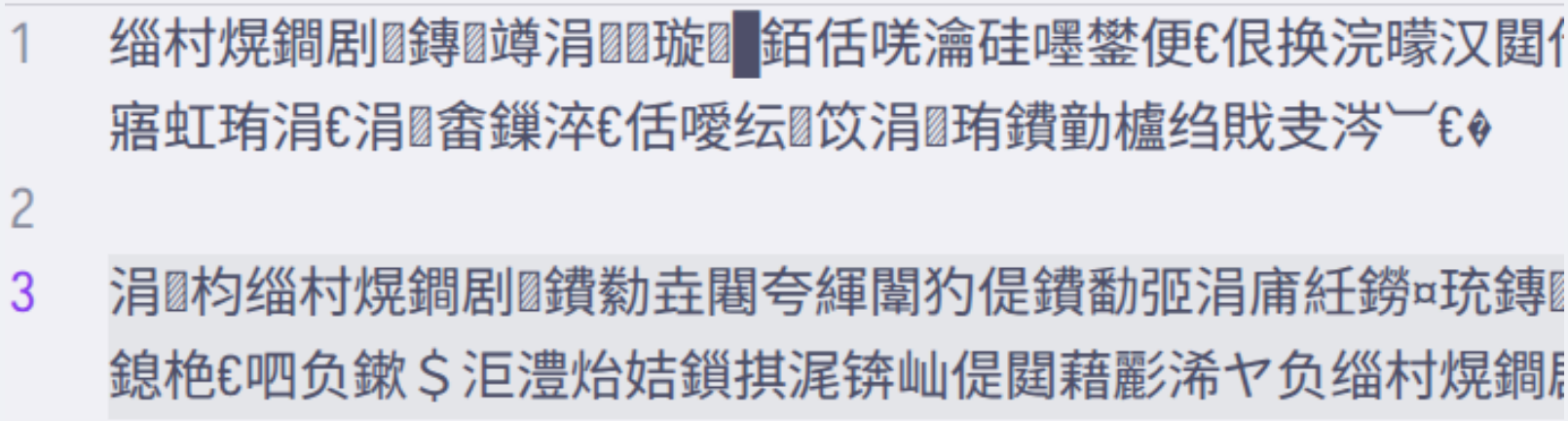
Kei Tang

2026-11-10

Recurse Center

Let’s first look at some CJK-related computer errors

1.1 Encoding errors



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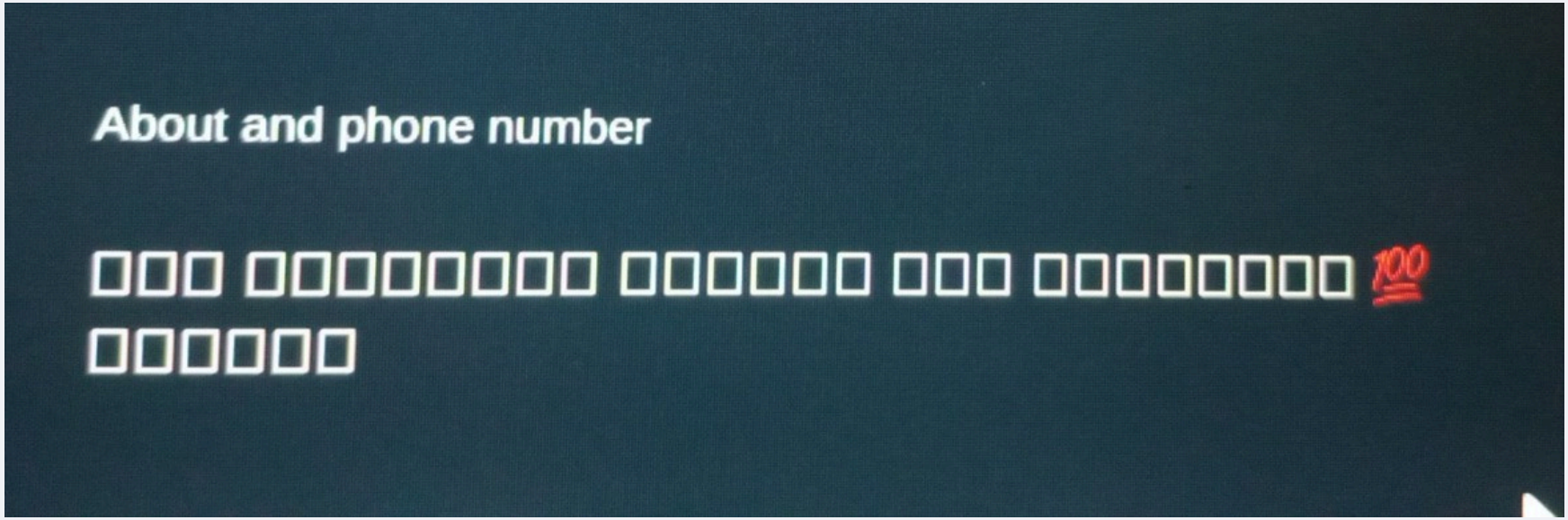


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[1]Cause: Reading UTF-8 as GBK/Reading GBK as UTF-8/Read GBK as UTF-8, save and read again as GBK

1.2 Font errors



[1]

[1] Missing CJK font (on linux)

1.2 Font errors



1

2

[1] Missing CJK font (on linux)

[2] Using Japanese font for Chinese, which has a limited set of characters

1.3 A bit of personal context

Unfortunately I have struggled with all of the above errors during my life tinkering with computers (╥﹏╥)

Why?

- My native language is (Simplified) Chinese
- I also read/write in Traditional Chinese and Japanese
- I want text from all those languages display correctly and beautifully:
  - on Linux systems and e-readers
  - on my personal website
  - on graphics software (eg. Illustrator)

CJK characters

Graphemes used in the **Chinese, Japanese, and Korean** writing systems, which each include Chinese characters (汉字/漢字).

Why are CJK fonts so complex (and sometimes annoying)?

2.1 A bit of regional history

Before the 20th century, almost all regions in eastern Asia use Traditional Chinese characters. In the 20th century, different governments made different changes to the writing system[1].

- **(Mainland) China**: In the 1950s, Simplified Chinese is legally standardized, outlawing the old Traditional characters[2].
- **Taiwan**: Traditional Chinese remains the standard.
- **Hong Kong/Macao**: Traditional Chinese remains popular, but with style differences.
- **Japan**: In 1946 and 1981, government published a limited set of standard Chinese characters, some of them are simplified.
- **Korea**: Since 1970s, Chinese characters are not part of the education, and rarely used in daily life.

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[1] Mostly to boost public literacy, but sometimes with political motives

[2] The same simplified character set is also the standard in Singapore and Malaysia.

2.2 CJK regional variants



2.3 Road to a unified character set

In the early days of computing, *different encoding standards* were proposed, making it very hard to share documents (and leading to the chaotic encoding errors).

- During the formation of the Unicode standard, **Han unification** was adopted. Different regional glyphs are generally considered to be the same “grapheme”, and assigned to a single unicode character[1].
- So, it’s now *the font’s* job to display those different regional variants!



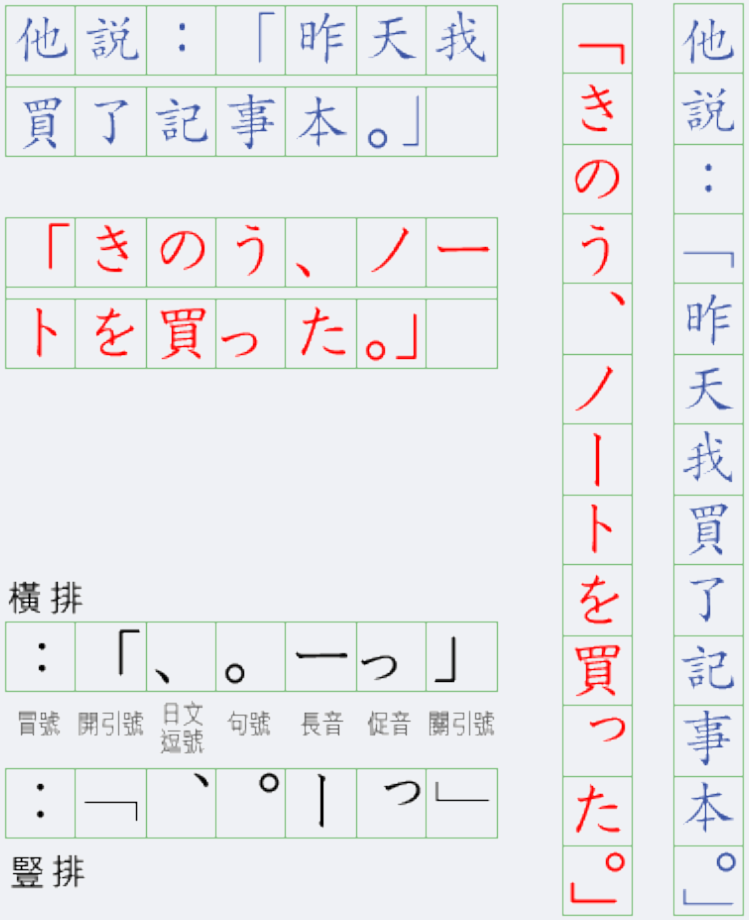
[1] However many characters were not unified due to compatibility reasons, and it’s complicated

Interlude:

Vertical or horizontal writing?

3.1 For CJK, it's complicated (again)

- Generally speaking, regions that use *Traditional* characters also use *vertical* writing (for books), including Taiwan and Japan.
- However, horizontal writing is becoming more common, because it's easier to mix with Latin words and has better software support.
- Since 2011, CSS has support for vertical writing, but almost all websites still use horizontal writing.



Now let’s talk about fonts

4.1 CJK font basics

Similar to English, there are several main typeface styles:

- Sans-serif: 黑体 (Simplified) / 黑體 (Traditional) / ゴシック体 (Japanese) ("Gothic" or "Black")
- Serif: 宋体/明體/明朝体 ("Song" or "Min")
- *Script/Handwritten*: 楷体/楷體 ("Kai")

Source Han Sans/Serif CJK[1] are the most popular open source CJK fonts, and they provide all regional variants.



[1] Also called Noto Sans/Serif CJK

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As a result, using software or browsing websites in a different CJK language variant is often broken or ugly (or both).

But, why???

5.1 There are just so many characters



- The latest Unicode standard (18.0) has **101,997** CJK characters.
- And we have multiple glyphs per character (due to Han unification), so the glyph number is much larger.
- Fortunately most of them is not really used. Source Han Serif CJK contains **65,535** glyphs, which is a small subset.
- Designing and building a font for so many glyphs is a lot of work.

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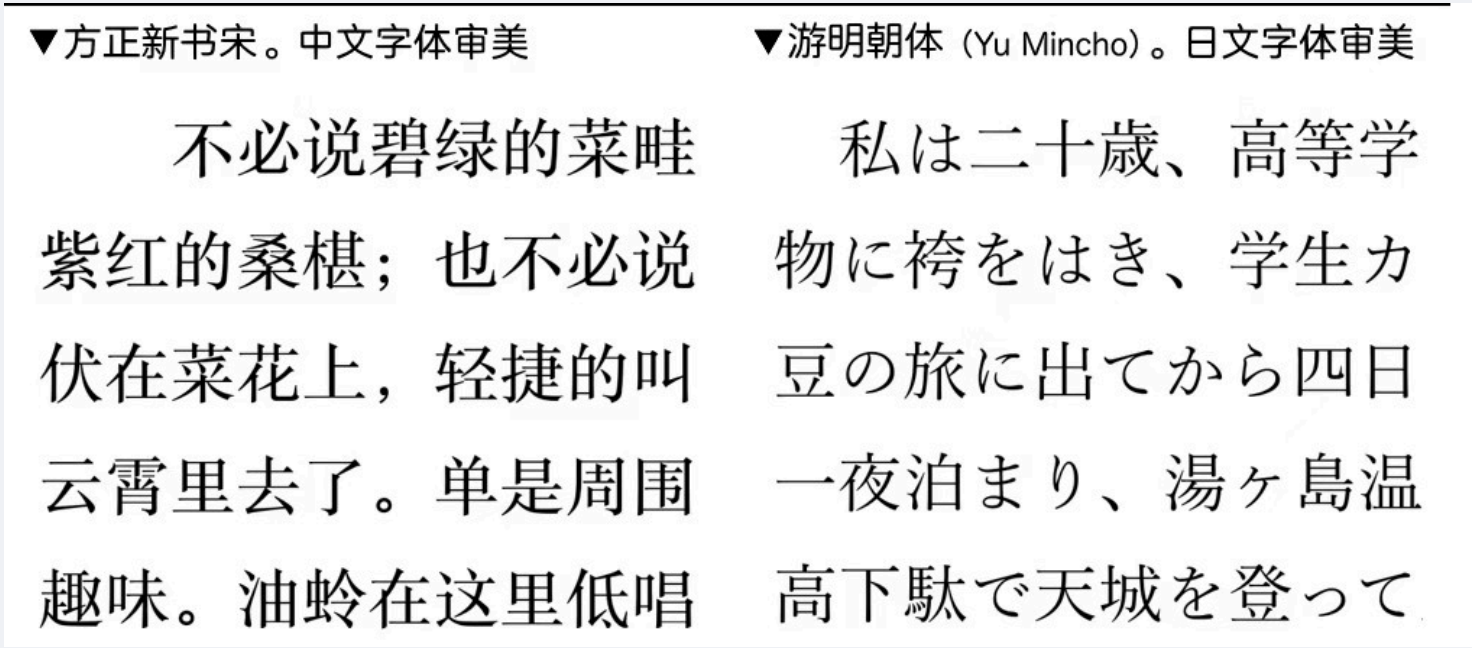
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Also,

- The file size for all region variants of Source Han Serif CJK is **over 100MB**.
- Imagine bundling it in your website (=_=)

5.2 Different languages require different design choices

Blindly transfering a font design to a different CJK language usually will not result in a good looking font.



Example: Well designed Chinese and Japanese Serif fonts

5.3 Case study: Serif fonts in Chinese and Japanese



Lines in Japanese fonts are usually thinner and softer

5.3 Case study: Serif fonts in Chinese and Japanese



In Chinese fonts, the visual shapes of characters are more “square”.

In Japanese fonts the shapes are more freeform.

5.3 Case study: Serif fonts in Chinese and Japanese

This softer, more flexible character design fits better in Japanese text, when mixed with *Kana*.



But when using the same style in Chinese text, it will appear messy and hard to read.

5.4 (Simplified) Chinese font scene is getting competitive



- The default Chinese fonts bundled with older versions of Windows[1] were terrible
- Stock Android use the Noto/Source Han font series, which is good but not great
- Since the 2020s, popular OS and phone vendors in China have released their own fonts, which have become an important part of their branding and marketing:
  - Apple: PingFang
  - Huawei: HarmonyOS Sans, Honor Sans
  - Oppo: Oppo Sans
  - Xiaomi: Mi Sans
  - Vivo: Vivo Sans
  - Oneplus: Oneplus Slate
  - ...

[1] before Windows 10

That’s all I have,

Hope you enjoyed my infodump on this rabbit hole :)